

Matgeo-4.4.15

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Question

Find the sum of the intercepts cut off by the plane
 $-(\mathbf{r}) \cdot (2\hat{i} + \hat{j} - \hat{k}) - 5 = 0$ on the three coordinate axes.

Solution

The given plane is

$$-(r) \cdot \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} - 5 = 0 \implies (r) \cdot \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} + 5 = 0 \quad (1)$$

X-intercept: $(x, 0, 0)$

$$(r) = \begin{pmatrix} x \\ 0 \\ 0 \end{pmatrix}, \quad (r) \cdot \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} + 5 = 0 \implies 2x + 5 = 0 \quad (2)$$

$$x = -\frac{5}{2} \quad (3)$$

Y-intercept: $(0, y, 0)$

$$(r) = \begin{pmatrix} 0 \\ y \\ 0 \end{pmatrix}, \quad (r) \cdot \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} + 5 = 0 \implies y + 5 = 0 \quad (4)$$

$$y = -5 \quad (5)$$

Solution(Continuation)

Z-intercept: $(0, 0, z)$

$$(r) = \begin{pmatrix} 0 \\ 0 \\ z \end{pmatrix}, \quad (r) \cdot \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} + 5 = 0 \implies -z + 5 = 0 \quad (6)$$

$$z = 5 \quad (7)$$

Conclusion

Conclusion: The intercepts cut off are $x = -\frac{5}{2}$, $y = -5$, $z = 5$.
Hence, the sum of intercepts is $-\frac{5}{2} - 5 + 5 = -\frac{5}{2}$.

Graphical Representation

